

NVision: A Sequential Monte Carlo Framework for Bayesian Experiment Design in Nitrogen-Vacancy Center Magnetometry

NVision Project¹

¹*Simulation framework described herein; physical implementation not performed.*

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Optically detected magnetic resonance (ODMR) of nitrogen-vacancy (N-V⁻) centers in diamond is a leading platform for nanoscale magnetometry, but conventional frequency-swept measurement wastes most of its budget sampling far from resonance. Sequential Bayesian experiment design (SBED) instead chooses each measurement to maximize expected information gain (EIG) about the resonance parameters [1]. This paper describes *NVision*, which extends this idea with a Sequential Monte Carlo (SMC) particle belief on a dimension-agnostic unit cube, scaling to Zeeman-split, hyperfine-split, and Voigt-broadened lineshapes with up to seven parameters. An EIG-maximizing acquisition rule, a closed-form Cramér–Rao lower bound (CRLB) per lineshape, and a multi-criterion stopping test jointly decide where to measure and when to stop. We present the signal models, the inference/acquisition mathematics, the simulation-harness architecture, and the evaluation metrics, all cross-checked against the reference implementation.

I. INTRODUCTION

Nitrogen-vacancy (N-V⁻) centers are diamond point defects with a spin-1 ground state whose $m_s = \pm 1$ sublevels Zeeman-split under an external field (Fig. 1); spin-dependent fluorescence makes this splitting optically readable as dips in an ODMR spectrum [2–4]. This makes N-V⁻ magnetometry a widely used nanoscale and vector field-sensing tool [5].

Conventional ODMR sweeps frequency over a fixed grid, wasting most measurements far from resonance where they barely change the fit uncertainty. Dushenko *et al.* [1] showed that choosing each frequency to maximize a posterior-based utility function instead — SBED — cuts the required measurements by more than an order of magnitude, using a two-parameter (frequency, linewidth) grid posterior and the **OptBayesExpt** library.

NVision generalizes this along three axes absent from that work and from an earlier internal HUII course attempt at the same problem [6] (Table I; bold = new here).

The particle-based unit-cube belief (Section III) is what makes the richer, multi-parameter lineshapes of Section II tractable in place of the grid both prior attempts share; the CRLB-derived stopping rule (Sections V–VI) is the paper’s most concrete methodological contribution, replacing a hand-tuned tolerance with one grounded in the measurement’s own information content. The EIG-maximization principle itself is Dushenko *et al.*’s [1, 7], and unlike that work — validated on real N-V⁻ hardware — NVision is simulation-only (Section VIII).

Section II covers the physical models and simulation architecture; Sections III–VI the inference/acquisition math; Section VII the evaluation metrics; Section VIII scope and limitations.

TABLE I. What NVision introduces relative to the two prior attempts. Bold = new here; plain = inherited.

Aspect	Prior attempts	This work
Posterior	OptBayesExpt grid, ≤ 2 –3 params	<i>N</i>-particle SMC, unit cube, up to 7
Signal models	Bare Lorentzian / one triplet	4 lineshape fami- lies (hyperfine, Zeeman, Voigt, saturation)
Stopping rule	Fixed tolerance / iteration count	Closed-form per- lineshape CRLB budget

II. PHYSICAL MODEL AND EXPERIMENT LAYOUT

A. N-V center structure and Zeeman splitting

An external field B along the N-V axis splits $m_s = \pm 1$ symmetrically about the zero-field frequency $f_B \approx 2.87$ GHz by $\gamma_{\text{NV}}B$ ($\gamma_{\text{NV}} \approx 28$ GHz/T), producing two Zeeman dips (Fig. 1); each is further split into a ¹⁴N hyperfine triplet (≈ 2.16 MHz) with finite homogeneous and inhomogeneous linewidth. Section II B’s lineshapes implement this hierarchy as composable, independently toggleable variants.

B. Signal models

All models are built around a single-Lorentzian-dip kernel

$$L(x; f, \Omega) = \frac{\Omega^2}{\Omega^2 + (x - f)^2}, \quad (1)$$

unit height at $x = f$, HWHM Ω (FWHM = 2Ω).

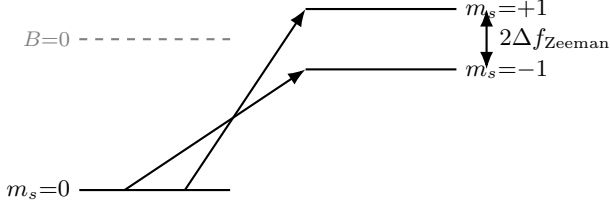


FIG. 1. NV^- ground-state spin triplet: field B splits $m_s = \pm 1$ symmetrically about $f_B \approx 2.87$ GHz by $\gamma_{\text{NV}}B$; microwave transitions from $m_s=0$ (arrows) give the two Zeeman-split ODMR dips.

$$p_{\text{sum}} = \frac{1}{k_{\text{np}}} + 1 + k_{\text{np}}, \quad p_0 = \frac{c}{p_{\text{sum}}}, \quad p_L = \frac{c}{k_{\text{np}} p_{\text{sum}}}, \quad p_R = \frac{c k_{\text{np}}}{p_{\text{sum}}}, \quad (p_L + p_0 + p_R \equiv c), \quad (2)$$

$$S(x) = 1 - \left[p_L L(x; f_0 - \Delta f, \Omega) + p_0 L(x; f_0, \Omega) + p_R L(x; f_0 + \Delta f, \Omega) \right], \quad (3)$$

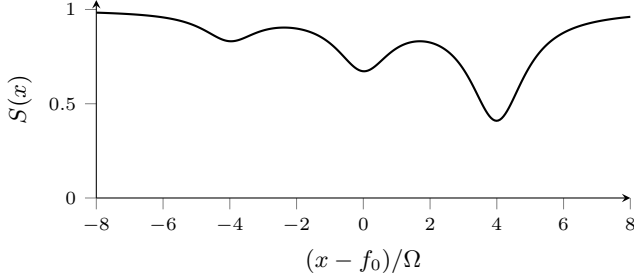


FIG. 2. Lorentzian triplet, Eq. (3), for $k_{\text{np}} = 2$, $\Delta f/\Omega = 4$: weights $p_L=c/7, p_0=2c/7, p_R=4c/7$ ($c=1$) sum exactly to c .

$k_{\text{np}} \in [1, 5]$. Unlike Eq. (1) of Dushenko *et al.* [1], where k_{np} scales two raw amplitudes without renormalizing, k_{np} here only redistributes c between dips. With hyperfine splitting off, Δf is fixed to 2.16 MHz and $k_{\text{np}} = 1$ (equal-weight triplet): the code calls this the “single dip” configuration, but Eq. (3) is still evaluated at nonzero splitting — only the number of *inferred* parameters drops (3 vs. 5 with hyperfine on), not the underlying triplet structure,

a. *Lorentzian triplet* (*NVCenterLorentzianModel*). With hyperfine splitting on, `nv_center_lorentzian_eval` uses a population-normalized reparameterization so the three dip weights always sum to the total contrast c (Fig. 2):

which visually blends into one dip only once $\Omega \gtrsim \Delta f$. With Zeeman splitting on, each dip weight in Eq. (2) is halved and the triplet duplicated at $f_0 \mp \Delta f_{\text{Zeeman}}$ (up to six free parameters, two dip clusters).

b. *Pseudo-Voigt triplet* (*NVCenterVoigtModel*). Each dip is a two-width pseudo-Voigt profile (`nv_center_pseudo_voigt_eval`), height-normalized as

$$V(x) = \frac{e_{\text{lf}}}{x^2 + \gamma_{\text{hom}}^2} + e_{\text{gf}} \exp\left(-\frac{x^2}{2\sigma_{\text{inhom}}^2}\right), \quad (4)$$

with $\gamma_{\text{hom}} = \frac{1}{2}\ell_{\text{frac}}\Gamma$, $\sigma_{\text{inhom}} = \frac{1}{2}(1-\ell_{\text{frac}})\Gamma/\sqrt{2\ln 2}$ splitting total FWHM Γ by Lorentzian fraction $\ell_{\text{frac}} \in [0, 1]$, and $e_{\text{lf}}, e_{\text{gf}}$ fixed via the Thompson–Cox–Hastings weight η so $V(0) = 1$ — algebraically the code’s on-the-fly normalization, pre-factored. This is not a true Voigt profile (no Faddeeva function), and η is calibrated for a shared FWHM rather than the split widths used here, so the true-Voigt approximation error is uncontrolled.

Unlike the Lorentzian family, this triplet is *not* population-normalized: `dip_depth` d sets the right-dip depth directly, with center and left dips at d/k_{np} and d/k_{np}^2 ,

$$S(x) = 1 - \left[\frac{d}{k_{\text{np}}^2} V(x - f_0 + \Delta f) + \frac{d}{k_{\text{np}}} V(x - f_0) + d V(x - f_0 - \Delta f) \right], \quad (5)$$

so total contrast is not held fixed as k_{np} varies, unlike Eq. (3). Population normalization reappears only in the saturation-coupled model below, which always routes through the Zeeman-population kernel ($\Delta f_{\text{Zeeman}}=0$

when Zeeman splitting is off).

c. *Saturation-coupled* (*NVCenterSaturationVoigtModel*). Linewidth and contrast are coupled through one drive parameter s

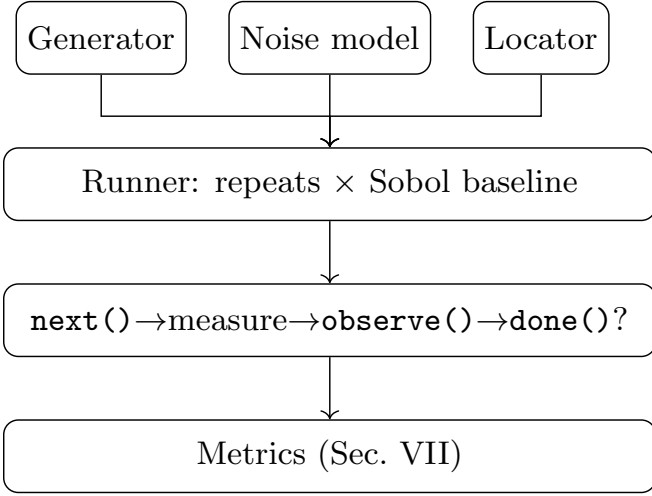


FIG. 3. Simulation harness: a Generator×Noise×Locator task runs under the Runner across independent repeats (Sec. IID), iterating until the stopping criteria of Sec. VI fire.

(power relative to half-saturation), since both are set by the same microwave field:

$$\gamma_{\text{hom}}(s) = \gamma_0 \sqrt{1+s}, \quad C(s) = c_{\text{max}} \frac{s}{1+s}, \quad (6)$$

with fixed zero-power HWHM $\gamma_0 = 150$ kHz and saturated contrast c_{max} . The independent Gaussian width σ_{inhom} is uncoupled to drive power. $(s, \sigma_{\text{inhom}}, c_{\text{max}})$ reparameterizes into $(\Gamma, \ell_{\text{frac}}, c)$ of Eq. (4) before evaluation; $\sigma_{\text{inhom}} \rightarrow 0$ recovers a pure Lorentzian exactly.

C. Physical parameter ranges

Table II lists the physical bounds shared by the signal generator and the SMC prior, so the two cannot drift apart.

$$\tilde{w}_i \leftarrow w_i p(y \mid \theta_i, x), \quad w_i \leftarrow \tilde{w}_i / \sum_j \tilde{w}_j, \quad \log p(y \mid \theta_i, x) = -\frac{1}{2} \left(\frac{y - S(x, \theta_i)}{\sigma} \right)^2, \quad (8)$$

where S is the signal at particle i ; the σ -only normalization constant cancels under renormalization. A tempering factor β rescales $\sigma \rightarrow \sigma/\sqrt{\beta}$; the update runs in

D. Simulation harness architecture

Each simulated experiment is a Generator×Noise×Locator task (Fig. 3): a ground-truth signal (Section IIB), an additive Gaussian noise floor (this paper’s scope; Poisson and Rao-Blackwellized inverse-gamma models exist but are not covered), and an acquisition-and-stopping strategy (uniform/Sobol baselines or a Bayesian locator). The Runner executes many tasks concurrently over independent, seed-deterministic, resumable repeats, streaming results to an on-disk cache for large repeat counts, and automatically runs a matched Sobol baseline for every Bayesian strategy under identical noise and ground truth. Within a repeat, `next()/observe()/done()` iterate until the stopping criteria of Section VI fire, recording the metrics of Section VII.

III. SEQUENTIAL MONTE CARLO BELIEF

The posterior over d free parameters θ is N weighted particles $\{(\theta_i, w_i)\}_{i=1}^N$, scaling to $d > 2$ without the grid’s exponential memory blow-up — the mechanism enabling Section IIB’s six-parameter models.

A. Unit-cube / physical coordinate separation

All particle arithmetic happens on a dimension-agnostic unit cube $[0, 1]^d$; each parameter j maps to its physical bound $[l_j, h_j]$ via a fixed affine rescale map,

$$\theta_j^{\text{phys}} = l_j + u_j (h_j - l_j), \quad \sigma_j^{\text{phys}} = \sigma_j^u (h_j - l_j), \quad (7)$$

owned exclusively by the belief and fixed for a run’s lifetime. This is kept strictly separate from the *focus window* (Section IVE) — the physical sub-interval the locator currently probes, which narrows over a run but never touches Eq. (7): the belief only sees the rescale map, acquisition/observation code only sees the focus window, preventing a class of coordinate-transform bugs.

B. Weight update

For observation (x, y) under Gaussian noise σ , each particle’s weight updates multiplicatively,

log-space with a max-shift to avoid underflow.

TABLE II. Representative physical parameter bounds (Lorentzian/Voigt family).

Parameter	Symbol	Range
Center frequency domain	f_0	2.6–3.1 GHz
Linewidth (HWHM)	Ω	200 kHz–5.0 MHz
Hyperfine splitting	Δf	2.0–8.5 MHz
Nuclear-polarization asymmetry	k_{np}	1.0–5.0
Zeeman half-separation	Δf_{Zeeman}	0–100 MHz ($\approx$ 0–3.6 mT)
Natural (zero-power) HWHM	γ_0	150 kHz (fixed)
^{14}N hyperfine constant	—	2.16 MHz (fixed, when not inferred)
Saturation parameter	s	0.02–30

C. Resampling

ESS = $1/\sum_i w_i^2$ triggers systematic resampling when ESS < $r_{\text{thr}}N$ ($r_{\text{thr}} = 0.20$): a single $U \sim \text{Uniform}(0, 1/N)$ gives positions $u_k = (k+U)/N$, mapped to particle indices by walking the weight CDF — lower variance than multinomial resampling.

D. Liu–West kernel

Post-resample particles contract toward the weighted mean $\boldsymbol{\mu}$ and jitter (Liu–West [8], with an exploration floor and rejuvenation):

$$\boldsymbol{\theta}_i \leftarrow a\boldsymbol{\theta}_i + (1-a)\boldsymbol{\mu} \quad (\text{shrinkage}), \quad (9)$$

$$\boldsymbol{\theta}_i \leftarrow \boldsymbol{\theta}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, (1-a^2)\boldsymbol{\Sigma}) \quad (\text{nudge}), \quad (10)$$

with $a = 0.98$, $\boldsymbol{\Sigma}$ the pre-resample covariance. The nudge covariance diagonal is floored,

$$C_{jj} \leftarrow \max\left(C_{jj}, [(h_j - l_j) f_{\text{expl}} e^{-t/25}]^2\right), \quad f_{\text{expl}} = 0.01, \quad (11)$$

so exploration decays but never vanishes (t = step count); eigen-regularization (min eigenvalue $\max(10^{-11}, 10^{-6}\lambda_{\text{max}})$) then guarantees a valid Cholesky

factor, and a decaying fraction $f_{\text{rejuv}} = 0.05 e^{-t/25}$ of particles are replaced by fresh prior draws to prevent mode collapse.

E. Weighted posterior statistics

Estimates and uncertainties are the weighted mean/std,

$$\mu_j = \frac{\sum_i w_i \theta_{ij}}{\sum_i w_i}, \quad \sigma_j = \sqrt{\frac{\sum_i w_i (\theta_{ij} - \mu_j)^2}{\sum_i w_i}}, \quad (12)$$

and the weighted covariance $\boldsymbol{\Sigma}$ feeds both the Liu–West kernel and the Gaussian entropy approximation

$$H \approx \frac{1}{2} \ln |\boldsymbol{\Sigma}| + \frac{d}{2} (1 + \ln 2\pi), \quad (13)$$

via a ridge-stabilized `slogdet`.

IV. SEQUENTIAL BAYESIAN EXPERIMENT DESIGN

A. Expected information gain

The acquisition rule approximates the EIG of candidate x by the particle cloud’s predictive variance,

$$\text{EIG}(x) \approx \frac{1}{2} \ln \left(1 + \frac{\sigma_{\text{pred}}^2(x)}{\sigma_{\text{noise}}^2} \right), \quad \sigma_{\text{pred}}^2(x) = \sum_i w_i S(x, \boldsymbol{\theta}_i)^2 - \left(\sum_i w_i S(x, \boldsymbol{\theta}_i) \right)^2, \quad (14)$$

with $\sigma_{\text{noise}}^2 = \max(\sigma^2, 10^{-12})$ — the exact entropy reduction for a Gaussian observation $y = S + \eta$, the same utility motivating SBED generally [1, 7]. Eq. (14) is evaluated on a stratified subsample of $\leq N_{\text{EIG}} = 500$ particles; between resamples the prediction matrix $M_{ci} = S(x_c, \boldsymbol{\theta}_i)$ is cached, so only $\sigma_{\text{pred}}^2 = M^{(2)}\mathbf{w} - (M\mathbf{w})^2$ is recomputed as weights change.

Candidates come from a slope-targeted epoch grid (dense windows at each dip’s $\pm\text{HWHM}$ points, plus a coarse global grid) rebuilt each resample. EIG scores are chunked (64 candidates); each chunk’s arg-max competes via a softmax at $\tau = 0.01$, avoiding lock-on to a single numerical-noise spike.

B. Exploration / dip-bias / EIG mixture

Each step, one uniform draw selects among three branches (Table III), so early search is not fully dependent on a possibly-biased posterior.

C. Background noise estimation

The noise std is not assumed known; it is estimated online from *background* measurements ($|x - \hat{f}| > k_{\text{bg}}|\hat{\Omega}|$, $k_{\text{bg}} = 3.0$) via the median absolute deviation,

$$\hat{\sigma}_{\text{bg}} = 1.4826 \text{ median } (|y_i^{\text{bg}} - \text{median}(y^{\text{bg}})|), \quad (15)$$

$$n_{\text{theory}} = \frac{4\hat{\sigma}_{\text{bg}}^2 \hat{\Omega} W}{\pi \hat{c}^2 T^2}, \quad N_{\text{budget}} = \max(N_{\text{max}}, K_{\text{theory}} n_{\text{theory}} + 1), \quad (16)$$

with $T = 100$ kHz, W the bandwidth, $K_{\text{theory}} = 20$ — large enough that this fires only when EIG acquisition has failed to converge far sooner.

E. Focus-window narrowing

Each resample narrows the search window to the union of particles' predicted active regions, $[f_i - \Delta f_i - k\Omega_i, f_i + \Delta f_i + k\Omega_i]$ ($k = 3.0$), using the 1st/99th-percentile edges to resist outliers; if particles pile against a unit-cube boundary ($>15\%$ within 5%), the window expands instead. This never mutates the fixed rescale map (Eq. 7).

V. FISHER INFORMATION AND THE CRAMÉR–RAO LOWER BOUND

A. General Gaussian Fisher information

For observation x under Gaussian noise, $\mathbf{I}(\boldsymbol{\theta}; x) = \sigma^{-2} \nabla_{\boldsymbol{\theta}} S \nabla_{\boldsymbol{\theta}} S^{\top}$; cumulative $\mathbf{I}_{\text{cum}} = \sum_{n=1}^N \mathbf{I}(\boldsymbol{\theta}; x_n)$, and $\text{CRLB}_j = \sqrt{[(\mathbf{I}_{\text{cum}} + \epsilon \mathbf{I})^+]_{jj}}$ ($\epsilon = 10^{-6}$) via a ridge-regularized Moore–Penrose pseudo-inverse [9].

B. Closed-form CRLB for a uniformly sampled Lorentzian

For $S = cL(x; f, \Omega)$ sampled at uniform density $\rho = N/W$,

$$\int_{-\infty}^{\infty} \left(\frac{\partial L}{\partial f} \right)^2 dx = \frac{\pi}{4\Omega}, \quad (17)$$

with $1.4826 = 1/\Phi^{-1}(3/4)$ the consistent-estimator factor. Below 15 background points, forced calibration samples the two background regions (weighted by width) until Eq. (15) is estimable.

D. Theoretical step-budget backstop

Once $\hat{\sigma}_{\text{bg}}$, $\hat{c} > 0$ are available, a permissive backstop (Section VB) bounds runaway,

so $I(f) = \rho c^2 \pi / (4\sigma^2 \Omega)$ and

$$\boxed{\text{Var}^{\text{CRLB}}(f) = \frac{4\sigma^2 \Omega}{\pi c^2 \rho}}, \quad \text{CRLB}_f = \sqrt{\text{Var}^{\text{CRLB}}(f)}, \quad (18)$$

uncertainty growing with σ, Ω and shrinking with c, ρ , as expected.

The same approach on the pseudo-Voigt (Eq. 4), dropping the Lorentzian–Gaussian cross-term (conservative: overestimates Var^{CRLB} for mixed lineshapes), gives

$$J = e_{\text{lf}}^2 \frac{\pi}{4\gamma_{\text{hom}}^5} + e_{\text{gf}}^2 \frac{\sqrt{\pi}}{2\sigma_{\text{inhom}}}, \quad \text{Var}^{\text{CRLB}}(f) = \frac{\sigma^2}{\rho c_{\text{total}}^2 J}, \quad (19)$$

reducing exactly to Eq. (18) as $\sigma_{\text{inhom}} \rightarrow 0$; both forms were checked against direct numerical integration of the coded kernels.

C. Numerical gradients revive the per-parameter FIM

None of the N-V[−] signal models expose an analytical $\nabla_{\boldsymbol{\theta}} S$, so CRLB_j (Eq. (18)'s general form, Section V) returned $\emptyset$ for every model in practice; the multi-parameter FIM machinery was present in the code but never exercised. A central-difference fallback, with step size scaled by each parameter's own bound range rather than its value (parameters here span ~ 7 orders of magnitude, from Hz-scale widths to a dimensionless contrast ≈ 0.25) and clamped into bounds, makes $\mathbf{I}_{\text{cum}}$ non-trivial for these models. Since the SMC belief and the FIM must share one coordinate system, the gradient point is taken in whatever coordinates the belief's model actually uses (unit-cube for `UnitCubeSMCMarginalDistribution`), with CRLB_j

TABLE III. Acquisition branch selection at step t .

Draw u	Action	Purpose
$u < 0.1 e^{-t/25}$	Uniform global sample	Decaying exploration
$u < 0.20$	Near empirical dip ± 5 MHz	Correct posterior bias
otherwise	Maximize EIG (Eq. 14)	Main information path

rescaled back to physical units by each parameter’s range afterward.

With the FIM live, non-frequency parameters in `reported_uncertainty()` are floored at their own marginal CRLB unconditionally (independent of the stopping-rule question addressed next): $\sigma_j^{\text{reported}} = \max(\sigma_j^{\text{phys}}, \text{CRLB}_j)$. Measured over 120 mixed repeats (56 deliberately degenerate), this moves median error/ σ^{reported} for `zeeman_split` from 1.21 to 0.90 and the fraction beyond 3σ from 8.3% to 2.5% (`c_total`: 1.61 $\rightarrow$ 0.96, 18.3% $\rightarrow$ 3.3%) — the particle spread alone understates uncertainty exactly on the near-flat ridge where `zeeman_split` trades off against the width pair, while the FIM correctly reports a large CRLB there because it is near-singular in that direction. No safety factor is applied, since the CRLB is already a hard lower bound; a $4\times$ factor (matching K_{safety}) was measured to

over-correct badly (medians drop to 0.09–0.25).

VI. CONVERGENCE CRITERIA

A run stops when *all* of the following hold for 8 consecutive resamples (`convergence_patience_steps`; any failure resets the streak, so a transient false positive cannot end a run early):

a. Per-parameter thresholds. Frequency: absolute $\sigma_f < T_f = 100$ kHz; others: relative 1% of bound width unless overridden. For saturation-Voigt, raw ($s, \sigma_{\text{inhom}}, c_{\text{max}}$) are not gated directly (inconsistent along the nonlinear saturation axis); instead their uncertainty propagates via the local Jacobian onto effective HWHM $\Omega(s, \sigma_{\text{inhom}}) = \gamma_0 \sqrt{1 + s} + \sqrt{2 \ln 2} \sigma_{\text{inhom}}$ and realized contrast $C(s, c_{\text{max}}) = c_{\text{max}} s / (1 + s)$, gated with ordinary linewidth/contrast semantics.

b. Overall RMS check. Even if every parameter passes individually,

$$u_j = \sigma_j / B_j, \quad B_j = \begin{cases} T_f / \text{threshold} & j = \text{frequency} \\ h_j - l_j & \text{otherwise} \end{cases}, \quad \text{RMS} = \sqrt{\frac{1}{d} \sum_j u_j^2} < \text{threshold}, \quad (20)$$

must also pass, so one marginal parameter cannot mask the rest.

c. Dynamic CRLB-derived stopping. Since $\text{CRLB}_f \propto 1/\sqrt{t}$, steps still needed are $n_{\text{req}} = t(\text{CRLB}_f / T_f)^2$; budget $N_{\text{dyn}} = \min(N_{\text{max}}, K_{\text{safety}} n_{\text{req}} + 1)$ ($K_{\text{safety}} = 4.0$), stopping as infeasible if $n_{\text{req}} > N_{\text{max}} K_{\text{safety}}$. This closed-form, frequency-only rule is unconditionally active. In SBED, the same idea extends to every parameter using the numerical-gradient FIM of Section VC, rescaled by the online noise estimate, $\text{CRLB}_j^{\text{scaled}} = \text{CRLB}_j^{\text{FIM}} \hat{\sigma}_{\text{bg}} / \sigma_{\text{nominal}}$, stopping when $\sigma_j < K_{\text{safety}} \text{CRLB}_j^{\text{scaled}}$ for every target parameter — but, as the next paragraph reports, this multi-parameter gate is kept *off* by default.

d. Why the multi-parameter FIM gate is disabled by default. Turning it on once Section VC’s fallback made CRLB_j non-trivial measurably improves the benchmark numbers — median run length 450 $\rightarrow$ 24 steps, catastrophic-error rate on known-degenerate configurations 67% $\rightarrow$ 12% — but we found this to be an artifact of the evaluation harness rather than a genuine capa-

bility. Near a degenerate point, the marginal CRLB is inflated by a near-singular $\mathbf{I}_{\text{cum}}$ (11.8 MHz measured on `zeeman_split` where the achieved error was only 1.3 MHz), so the stopping test passes trivially from the first few steps — exactly where the problem is hardest. The apparent accuracy gain traces to the generators’ own prior: each repeat’s SMC prior mean is drawn as $\mathcal{N}(\theta_{\text{true}}, \sigma_{\text{prior}})$, so stopping early scores well because the run reports a truth-centred prior it never had to earn against real measurements. This is the same failure class as three earlier snapshot-CRLB early-stop bugs found during development of this locator, and we flag it here as a caution against validating simulation-based stopping rules purely against simulated-prior-centred accuracy.

e. Adaptive plateau stopping (default). In place of the FIM gate, the shipped default stops a run once every target parameter’s *point estimate* has stopped moving relative to its own current uncertainty — measuring diminishing returns directly rather than trusting a derived quantity’s claim about the information limit. Writing

$\hat{\theta}_j^{(t)}$ for the estimate at check t ,

$$\text{plateaued}_j \iff |\hat{\theta}_j^{(t)} - \hat{\theta}_j^{(t-W)}| < f_\sigma \sigma_j^{(t)}, \quad (21)$$

with window $W = 30$ checks and fraction $f_\sigma = 0.25$; a run stops once every target parameter plateaus for `convergence_patience_steps` consecutive checks. Movement is scaled by each parameter’s own σ (not its bound range), so one threshold applies across parameters spanning orders of magnitude. On the N-V Voigt grid (120 repeats), median `zeeman_split` error falls 1.483 MHz (step 10) $\rightarrow$ 0.237 (25) $\rightarrow$ 0.059 (50) $\rightarrow$ 0.022 (100) $\rightarrow$ 0.016 (450) MHz — a knee near step 100, after which the remaining 350 of a 450-step budget buy $\sim 1.3\times$ accuracy for $4.5\times$ the measurements. $f_\sigma = 0.25$, calibrated by replaying the rule inside full-budget runs and confirmed with a live A/B (120 configs/arm), fires on 100% of both ordinary and degenerate configurations, saving $2.3\times$ and $3.6\times$ measurements respectively (median step 450 $\rightarrow$ 197 ordinary, 450 $\rightarrow$ 124 degenerate) while accuracy on most target parameters is within a few percent of the full-budget run — well under the $\sqrt{2}$ – $\sqrt{3.6}$ increase a naive $\sqrt{n}$ scaling would predict. The one measured cost: on degenerate configurations the catastrophic-error rate rises 7.1% $\rightarrow$ 12.5% (4 $\rightarrow$ 7 of 56 repeats); a more conservative $f_\sigma = 0.15$ keeps most of that margin at a $2.2\times$ rather than $3.6\times$ saving.

Table IV summarizes the defaults above.

Novelty. The Fisher/Cramér–Rao construction is textbook [9]; new here are (i) closed-form CRLBs for this lineshape family (Eqs. 18–19), absent from either prior attempt or Dushenko *et al.* [1], used as the frequency-only stopping/feasibility mechanism in place of fixed-iteration or fixed-tolerance rules — arguably the paper’s most concrete methodological contribution — and (ii) the negative result that the natural multi-parameter extension of the same idea (gate every parameter on its own FIM-derived CRLB) is not safe to use as-is under a benchmark with an informative simulated prior, together with the adaptive plateau rule we use instead, which achieves a comparable 2–3.6 $\times$ measurement reduction without that failure mode.

VII. EVALUATION METRICS

A. Point-estimate error

For a single peak, $\text{abs_err}_x = |\hat{f} - f_{\text{true}}|$; for two peaks (e.g. a resolved Zeeman doublet), estimates and truths are sorted and paired before computing $\text{pair_rmse} = \sqrt{\frac{1}{2}(\text{abs_err}_{x_1}^2 + \text{abs_err}_{x_2}^2)}$.

B. Uniform-sweep baseline

To quantify savings, each run is compared against the uniform measurement count needed for the *same* ground

truth. Nyquist argument: a feature of width w is resolvable at spacing $\delta \leq w/2$, giving

$$n_{\text{sweep}} = \frac{2W}{w_{\text{eff}}}, \quad (22)$$

with w_{eff} measured on the true signal (20,000-point grid, 20%-depth threshold, segments merged within 2% of the domain) rather than a worst-case bound. E.g. $W = 500$ MHz, $\Omega \approx 100$ kHz ($w = 4\Omega \approx 400$ kHz) gives $n_{\text{sweep}} \approx 2500$ — scaling as W/Ω , not fixed. Reported $\text{sobel_difference} = n_{\text{sweep}} - n_{\text{measured}} > 0$ whenever SBED used fewer measurements.

C. Failure classification

Each run gets one failure reason, in priority order: success (frequency converged, or an unconverging baseline); `infeasible_crlb` (pre-run gate tripped); `timeout`; `theory_budget` (Section IV count exceeded); `max_steps` (budget exhausted otherwise). The `infeasible_crlb` gate reuses `marginal_crlbs_at_budget`, which — like `crlb_per_param()` before Section VC’s fallback — requires an analytical gradient and was not extended to the numerical fallback; for every N-V⁺ model in this paper’s scope it therefore returns no CRLBs and this gate never actually fires, so `infeasible_crlb` does not occur among the results reported here.

VIII. DISCUSSION

NVision keeps Dushenko *et al.*’s central claim — EIG maximization converges in far fewer measurements than a fixed sweep — but swaps the two-parameter grid for an N -particle SMC belief, trading grid intractability for particle noise at small N (mitigated by Section III’s resampling/shrinkage machinery and the CRLB backstop of Sections V–VI).

Two approximations bound these claims: the pseudo-Voigt (Eq. 4) is not a true Voigt profile, with uncontrolled deviation from one; and the Voigt CRLB (Eq. 19) drops the Lorentzian–Gaussian cross-term, conservative but inexact for strongly mixed lineshapes. Both are checked numerically against the coded kernels, not an independent analytic Voigt CRLB. This paper’s scope is the additive Gaussian noise path; Poisson and Rao–Blackwellized inverse-gamma noise models exist in the code but are not covered. Finally, metrics are computed against the true signal’s resolvable width rather than a worst-case parameter bound, since the same Ω range (200 kHz–5 MHz) spans an order of magnitude in sweep budget.

A further caveat concerns any stopping-rule accuracy claim measured on this harness: because each repeat’s SMC prior is drawn centred on that repeat’s own ground truth (Section III), a rule that stops as soon as some

TABLE IV. Key inference-stack constants and their defaults.

Symbol	Meaning	Default
N	SMC particle count	1000
r_{thr}	ESS resampling threshold (fraction of N)	0.20
a	Liu–West shrinkage	0.98
N_{EIG}	EIG subsample size	500
T_f	Frequency convergence threshold	100 kHz
K_{safety}	CRLB safety factor	4.0
K_{theory}	Theory-budget safety factor	20
k_{bg}	Background classification span (linewidths)	3.0
patience	Consecutive converged resamples required	8
threshold	Relative convergence threshold	1%
W	Plateau-stop window (checks)	30
f_{σ}	Plateau-stop drift fraction	0.25

derived quantity looks converged can score well by reporting a prior it never had to earn, not by having actually resolved the signal — Section VI’s FIM-gate finding is one concrete instance of this, and it should be treated as a general risk when validating any stopping criterion, CRLB- or otherwise, purely against simulated-prior-centred error.

IX. CONCLUSION

NVision generalizes single-dip SBED for N-V⁻ ODMR [1] to hyperfine, Zeeman, Voigt, and saturation-coupled lineshapes via a dimension-agnostic SMC belief. An EIG acquisition rule, online noise estimation, and per-lineshape closed-form CRLBs jointly decide where to measure and when to stop; the accompanying harness runs this at scale against Nyquist-derived uniform-sweep and Sobol baselines on matched ground truth, with derivations cross-checked against the reference implementation throughout.

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